

shares, "RBEI started 20 years ago with design for European, USA, Japanese and Korean customers. Eighty per cent of our work is still for these customers. Meanwhile, BRIC (Brazil, Russia, India and China) has been our special focus. We are responsible for doing platforms for these OEMs."

Target application areas

Indian IDHs cater to a wide range of industry segments and verticals.

"Being a large country with ample human resource, India offers enough business opportunities for IDHs. You will find design houses catering to needs in product design and development (including electronics, appliances, entertainment, packaging and machinery), application design and development (applications covering a wide array of Web and mobile applications, enterprise applications, utility applications, embedded applications, lab and hospital information systems, etc) and new product innovation," informs Nallan.

"Most of the IDHs work as offshore design centres for some of the major Tier 1 OEMs. Indian IDHs are now transforming from being just a product sustenance/maintenance provider to complete product inception to design to production partners. IDHs in India are involved from architecting the product to market rollout and sustenance," adds Panapakam.

"The target application areas are semiconductor (ASIC design, hardware design, firmware/embedded software and middleware), automotive, telecom and networking, medical electronics and industrial automation. Major strengths are in the areas of ASIC design, hardware design, embedded software and vertical specific software design and development," he shares.

However, areas differ from company to company. For example, "Calixto Systems is focusing on the Internet of Things (IoT) and microprocessor-based designs. There are a lot of companies in India working on multimedia-based designs," says Jithu Niruthambath, director of Calixto Systems.

Strategies for growth

1. Mindset change on working with customers as 'partners' rather than being just service providers
2. Collaboration between OEMs, Tier-x suppliers, government bodies and consumers to accelerate the development and introduction of newer technologies
3. Changing gears from volume growth to value growth, along with productivity focus to ensure long-term sustenance
4. Increased 'product domain' competency by enhancing university cooperation and research at institutes
5. Promoting cross-domain working in matrix teams and fostering innovation alongside an environment that promotes diversity in the workplace (across gender, age, region to name a few)

— **R.K. Shenoy, senior vice president,
Powertrain Electronics, Robert Bosch
Engineering & Business Solutions**

Specifically talking about RBEI, Shenoy shares, "We are working here in automotive, industrial, consumer and energy domains. Our strength has been working on complex mechatronic systems (software, hardware and mechanics), with close proximity to customers, sales/marketing, manufacturing and purchasing. Working for the Indian market as end users has also been one of our biggest strengths as we graduated from working on extended workbench to take on full responsibility projects with focus on innovation. In areas like low-priced vehicles, we engineer our products with 'frugal engineering/jugaad engineering.' I would like to stress that working on low-cost systems doesn't mean low technology. Rather, it is cost innovation to get the appropriate technology to the market."

Challenges and threats faced

"As IDHs move up the value chain and start doing complete product designs, one of the major challenges faced is the availability of qualified and experienced manpower. Also, many a times IDHs lose these qualified and experienced engineers to captive units of OEMs/MNCs operating out of India.

The option here is to provide generous stock options and ensure that the engineers are continuously occupied with interesting work," says Panapakam.

"Conducive regulatory environment is important for service-driven companies to grow. A startup survives through all hassles because of the sheer energy. However, once a company reaches maturity, it needs to cope up with a lot of environmental issues. Due to policies, Indian companies are compelled to undertake a lot of processes, compliances and paperwork leading to redundancy and frustration. Mid-size companies have to cope up with talent crunch while achieving scale and growth," adds Nallan.

Talking of threats, Nallan adds, "Indian design houses do face threat from their foreign counterparts and this is going to only increase in the future. Substitutes also pose a threat: One-stop shops like China and Taiwan can lure companies into ready solutions and thus not investing in design activities."

"IDHs are usually small in size and require high-calibre engineers. The main issues we have come across are hiring good resources and retaining them due to competition, and R&D funding," shares Niruthambath.

Strategies adopted by IDHs to foster growth

"IDHs need to identify and work with not just large OEMs but also mid-tier and interesting startups. Many a times, new ideas and products come out of startups and it would be in the interest of IDHs to align with these companies to foster growth. Another advantage of working with these mid-tier and startups would be reduced selling time and less red tapism as compared to working with large corporations. However, working with startups requires innovative business practices and also involves a certain amount of risk," says Panapakam.

According to Nallan, keys to growth include "talent acquisition, training and development, and talent retention; offering quality services at affordable cost; building formidable